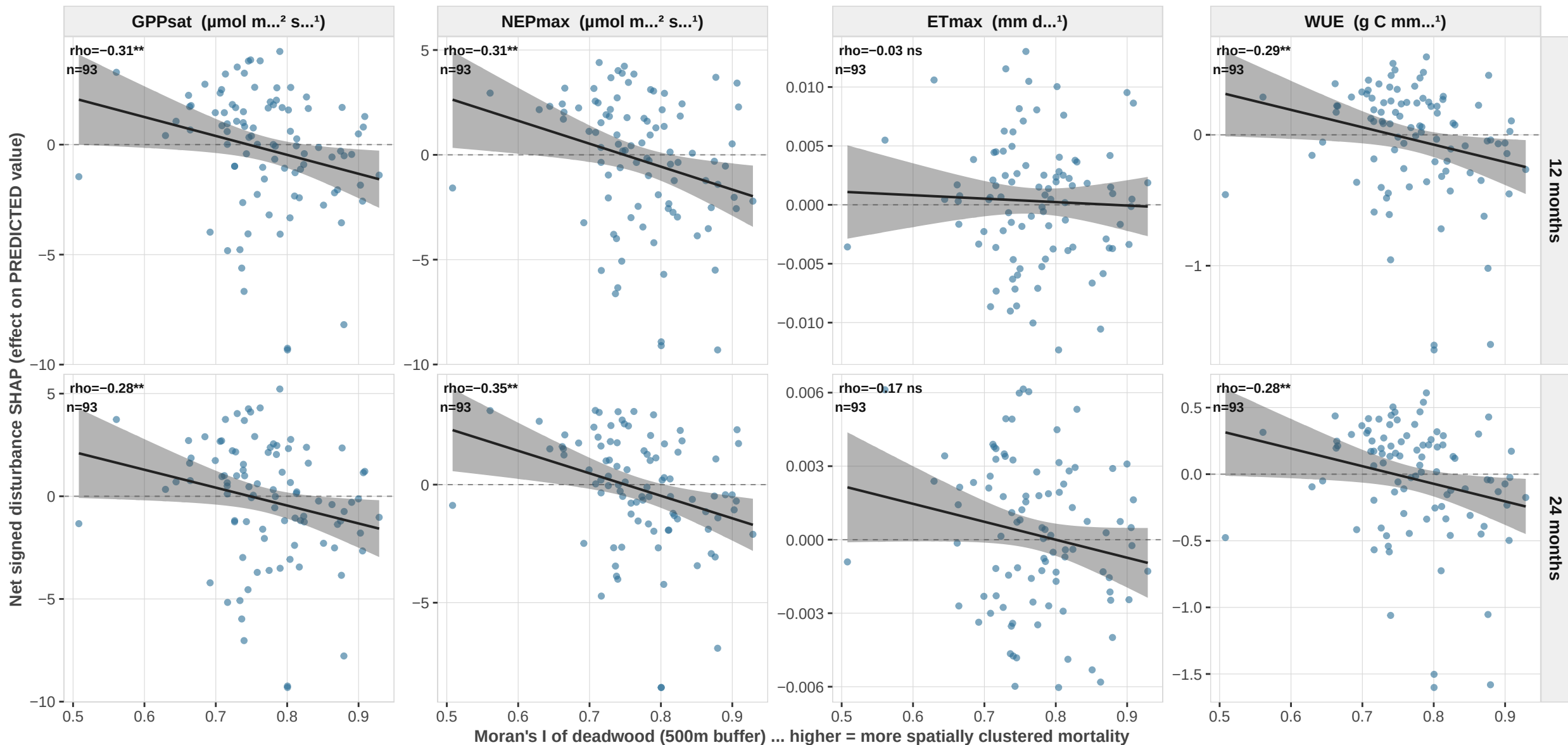


# Does spatial clustering of mortality predict the direction of the disturbance effect?

Moran's I (deadwood layer, rook contiguity, 500m circular buffer, site-level mean across years) vs net signed disturbance SHAP  
tree cover >= 30% (93 sites) | XGBoost-Optuna, M4 | linear fit with 95% CI | Spearman rho, BH-FDR across all 8 tests



Below zero = model predicts a LOWER EFP value where mortality is present (not worse accuracy, see Fig 4).